

EEG forward operator with a template MRI

This tutorial explains how to compute the forward operator from EEG data using the standard template MRI subject [fsaverage](#).

⚠ Caution

Source reconstruction without an individual T1 MRI from the subject will be less accurate. Do not over interpret activity locations which can be off by multiple centimeters.

Adult template MRI (fsaverage)

First we show how `fsaverage` can be used as a surrogate subject.

```
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#
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import numpy as np

import mne
from mne.datasets import eegbci, fetch_fsaverage

# Download fsaverage files
fs_dir = fetch_fsaverage(verbose=True)
subjects_dir = fs_dir.parent

# The files live in:
subject = "fsaverage"
trans = "fsaverage" # MNE has a built-in fsaverage transformation
src = fs_dir / "bem" / "fsaverage-ico-5-src.fif"
bem = fs_dir / "bem" / "fsaverage-5120-5120-5120-bem-sol.fif"
```



Out:

```
0 files missing from root.txt in /home/circleci/mne_data/MNE-fsaverage-data
0 files missing from bem.txt in /home/circleci/mne_data/MNE-fsaverage-data/fsaverag
```

Load the data

We use here EEG data from the BCI dataset.

Note

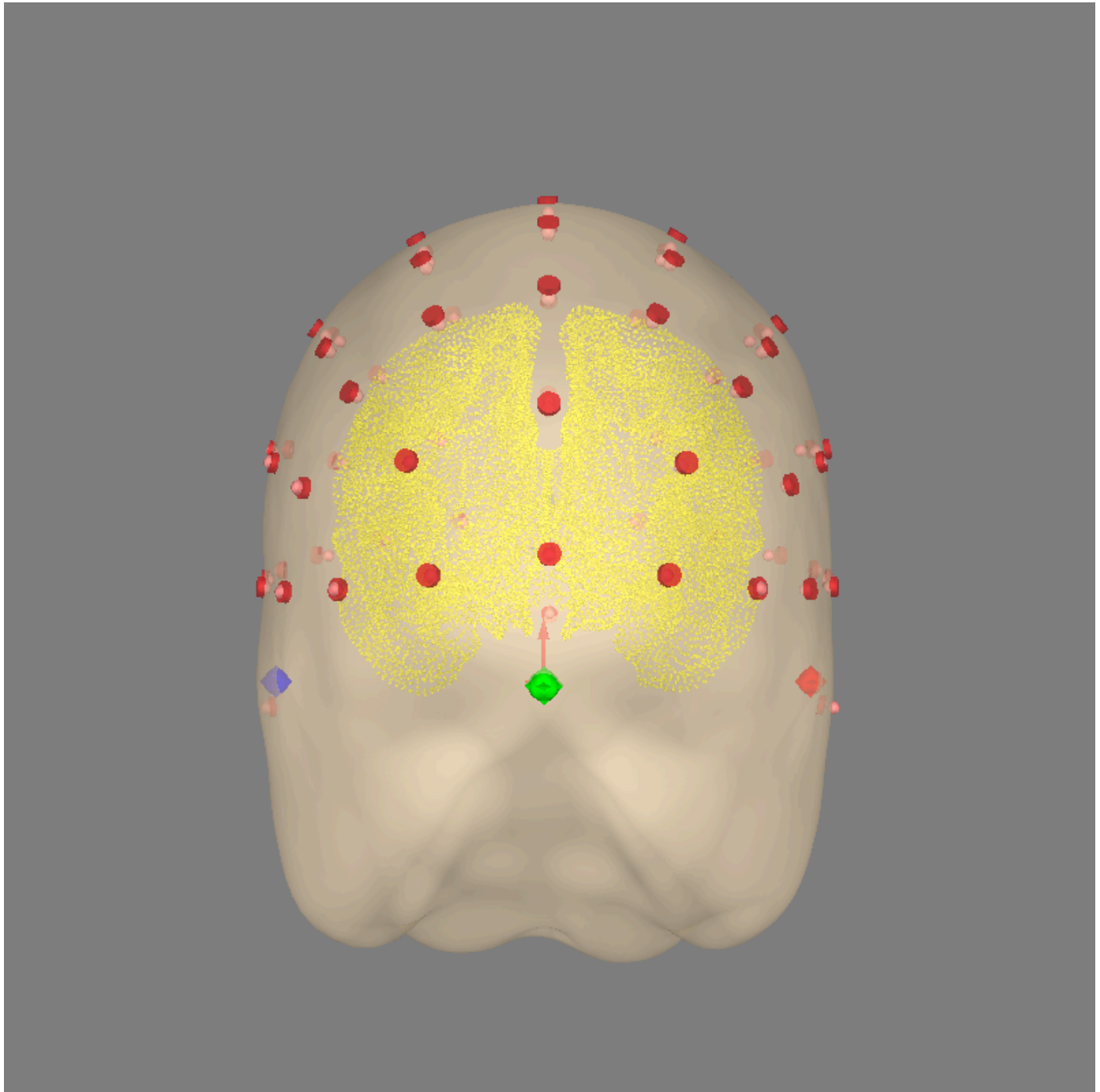
See [Plotting sensor layouts of EEG systems](#) to view all the standard EEG montages available in MNE-Python.

```
(raw_fname,) = eegbci.load\_data(subjects=1, runs=[6])
raw = mne.io.read\_raw\_edf(raw_fname, preload=True)

# Clean channel names to be able to use a standard 1005 montage
eegbci.standardize(raw)

# Read and set the EEG electrode locations, which are already in fsaverage's
# space (MNI space) for standard_1020:
montage = mne.channels.make\_standard\_montage("standard_1005")
raw.set_montage(montage)
raw.set_eeg_reference(projection=True) # needed for inverse modeling

# Check that the locations of EEG electrodes is correct with respect to MRI
mne.viz.plot\_alignment(
    raw.info,
    src=src,
    eeg=["original", "projected"],
    trans=trans,
    show_axes=True,
    mri_fiducials=True,
    dig="fiducials",
)
```



Out:

```
Source space      : /home/circleci/mne_data/MNE-fsaverage-data/fsaverage/bem/fsa
MRI → head transform : /home/circleci/project/mne/data/fsaverage/fsaverage-trans.f
Measurement data   : instance of Info
Conductor model    : /home/circleci/mne_data/MNE-fsaverage-data/fsaverage/bem/fsaver
Accurate field computations
Do computations in head coordinates
Free source orientations
```

```
Reading /home/circleci/mne_data/MNE-fsaverage-data/fsaverage/bem/fsaverage-ico-5-sr
Read 2 source spaces a total of 20484 active source locations
```

```
Coordinate transformation: MRI (surface RAS) → head
```

```
0.999994 0.003552 0.000202    -1.76 mm
-0.003558 0.998389 0.056626    31.09 mm
-0.000001 -0.056626 0.998395    39.60 mm
0.000000 0.000000 0.000000     1.00
```

```
Read 64 EEG channels from info
Head coordinate coil definitions created.
Source spaces are now in head coordinates.
```

```
Setting up the BEM model using /home/circleci/mne_data/MNE-fsaverage-data/fsaverage
```

```
Loading surfaces ...
```

```
Loading the solution matrix ...
```

```
Three-layer model surfaces loaded.
```

```
Loaded linear collocation BEM solution from /home/circleci/mne_data/MNE-fsaverage-d
```

```
Employing the head→MRI coordinate transform with the BEM model.
```

```
BEM model fsaverage-5120-5120-5120-bem-sol.fif is now set up
```

```
Source spaces are in head coordinates.
```

```
Checking that the sources are inside the surface and at least 5.0 mm away (will
```

```
Checking surface interior status for 10242 points ...
```

```
Found 2433/10242 points inside an interior sphere of radius 47.7 mm
Found 0/10242 points outside an exterior sphere of radius 98.3 mm
Found 0/ 7809 points outside using surface Qhull
Found 0/ 7809 points outside using solid angles
Total 10242/10242 points inside the surface
```

```
Interior check completed in 7810.7 ms
```

```
Checking surface interior status for 10242 points ...
```

```
Found 2241/10242 points inside an interior sphere of radius 47.7 mm
Found 0/10242 points outside an exterior sphere of radius 98.3 mm
Found 0/ 8001 points outside using surface Qhull
Found 0/ 8001 points outside using solid angles
Total 10242/10242 points inside the surface
```

```
Interior check completed in 7526.8 ms
```

```
Setting up for EEG ...
```

```
Computing EEG at 20484 source locations (free orientations) ...
```

```
Finished.
```

Forward		
EEG	64	
Source space	Surface with 20484 vertices	
Source orientation	Free	

From here on, standard inverse imaging methods can be used!

Infant MRI surrogates

We don't have a sample infant dataset for MNE, so let's fake a 10-20 one:

```
ch_names = "Fz Cz Pz Oz Fp1 Fp2 F3 F4 F7 F8 C3 C4 T7 T8 P3 P4 P7 P8 O1 O2".split()
data = np.random.RandomState(0).randn(len(ch_names), 1000)
info = mne.create_info(ch_names, 1000.0, "eeg")
raw = mne.io.RawArray(data, info)
```

Out:

```
Creating RawArray with float64 data, n_channels=20, n_times=1000
Range : 0 ... 999 = 0.000 ... 0.999 secs
Ready.
```

Get an infant MRI template

To use an infant head model for M/EEG data, you can use

`mne.datasets.fetch_infant_template()` to download an infant template:

```
subject = mne.datasets.fetch_infant_template("6mo", subjects_dir, verbose=True)
```

Out:

```
0 files missing from ANTS6-0Months3T.txt in /home/circleci/mne_data/MNE-fsaverage
```

It comes with several helpful built-in files, including a 10-20 montage in the MRI coordinate frame, which can be used to compute the MRI $\leftrightarrow$ head transform `trans`:

```
from mne.io import read_template
montage = read_template('10-20_montage_512.elc')
```

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Built with the [PyData Sphinx Theme](#)

GMT+3.

0.17.1.